

GAINS: A Meta-Reinforcement Learning Model for Gated Arbitration of Individual and Social Learning Signals

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Abstract—Social learning enables individuals to improve decision making by using information generated by others to update their own beliefs. In humans, social learning is known to be selective, depending on factors such as demonstrator expertise, yet social information can be informative even when the sources are unreliable. We introduce GAINS (Gated Arbitration of Individual and Social Learning Signals), a meta-reinforcement learning architecture designed for social learning in a bandit task setting. GAINS learns to adaptively arbitrate between self-experienced and socially observed evidence under realistic conditions in which both demonstrator reliability and task difficulty are uncertain. GAINS is evaluated across three information regimes derived from a two-choice visual bandit task: Individual Learning (IL), relying solely on self-experience; Action-Only (AO), providing partial access to another agent's actions; and Observational Learning (OL), offering full access to both actions and outcomes. We use GAINS to examine how different demonstrator reliability regimes and task difficulty levels shape social learning. Finally, from a developmental perspective, we manipulate memory accessibility in GAINS to investigate how access to past experience influences social learning performance.

I. INTRODUCTION

Humans are efficient social learners: rather than relying solely on direct experience, they observe others, infer useful information from their behavior, and selectively adopt behaviors worth emulating. This process is not mere passive imitation; it often requires complex computations, including reasoning under partial observability and dynamically weighing social evidence against self-experience. Computational accounts of social learning have begun to formalize these processes, modeling how individuals extract information from others' actions and outcomes [1], [2], infer demonstrator competence [3], and arbitrate between information sources [4]. Yet most existing approaches either encode social influence through fixed weights [5], [6] or rely on demonstrations that are more informative than those available in naturalistic settings [7], [8], rather than learning to evaluate observational evidence under noise and uncertainty about demonstrator reliability and task rewards. In humans, social

learning is often selective, influenced by factors such as motivation and perceived reliability of others [9]. In developmental robotics, related work has explored how agents evaluate and select interaction partners based on internal signals such as cognitive load [10]. Building on these ideas, we investigate how agents can adaptively learn from available social and individual information.

We introduce GAINS (Gated Arbitration of Individual and Social Learning Signals), a computational framework for adaptive social learning. We formulate GAINS as a meta-reinforcement learning agent [11] that discovers how to integrate self-experience with observational evidence in a context-sensitive manner. Unlike classic machine learning approaches such as imitation learning [12], [13] or inverse reinforcement learning [14], GAINS does not assume access to expert demonstrations, nor does it aim to recover a demonstrator's policy [8], reward function [15], or intent [16], [17]. Instead, it addresses a more general question: how can an agent use socially available information to improve its own decisions under uncertainty?

This question is particularly important because, in social learning contexts, even suboptimal or failed demonstrations can provide useful information. Yet human learners often rely on trust-based heuristics, such as perceived expertise [18], [19] when weighing social evidence, which may lead them to discount potentially informative but unreliable sources. By contrast, GAINS learns how to arbitrate between social and self-generated evidence in such a way that social information can be utilized effectively to maximize returns. It is also related to Theory-of-Mind approaches such as ToMNet [20], which infer latent properties of other agents from observed behavior to support decision-making. In addition, GAINS incorporates a memory-augmented architecture [21], [22], enabling rapid storage and retrieval of task-relevant information to support flexible learning.

We evaluate GAINS using a series of experiments based on the two-armed bandit paradigm developed to study social learning in humans [23], [24]. We extend this paradigm by systematically manipulating key factors, including demonstrator reliability, expertise, and task difficulty. This framework enables us to examine how agents weight social evidence under a range of conditions, such as exposure to unreliable or expert demonstrators, non-stationary policies, and partial observability. Finally, from a developmental perspective, we manipulate memory accessibility to examine how limiting access to past experiences shapes the arbi-

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Condition	Stimulus Pair	Choice → Outcome
IL		→ → +1
AO-o		→ → → ?
AO-s		→ → +1
OL-o		→ → → +1
OL-s		→ → 0

Fig. 1. Two-Choice Visual Bandit Task: We use two-choice bandit task with 5 different conditions for investigating social learning. For each session, the environment generates three novel image pairs using a custom generator that creates images composed of randomly arranged geometric shapes and colors. **IL** trials involve direct self-experience only. **AO-o** and **OL-o** are observation trials in which GAINS watches another agent choose, with reward hidden in AO-o and visible in OL-o. **AO-s** and **OL-s** are later self-choice trials on the same stimulus pairs, used to assess how previously observed social information influences GAINS’s own decisions

tration between social and individual learning. We use this manipulation as a computational hypothesis about a possible mechanism underlying developmental differences in social learning [25], without attempting a direct comparison with human data in the present study.

II. METHODS

A. Two-Choice Visual Bandit Task

We study social and individual learning in a two-choice visual bandit task across multiple sessions. In each session, new visual stimuli are introduced, requiring the agent to learn an internal reinforcement-learning-like strategy to identify the option that yields higher rewards in that session. Each bandit ($i \in \{1, 2\}$) has an unknown reward probability ($p_i \in [0, 1]$), fixed within a session of (T) trials. On trial (t), the agent observes images ($x_{t,L}, x_{t,R}$), selects ($a_t \in \{L, R\}$), and receives a binary reward ($r_t \in \{0, 1\}$) based on the chosen bandit.

Trials are organized into five conditions as shown in Fig. 1: individual learning (IL), action-only observation with self and observation phases (AO-s, AO-o), and observational learning with self and observation phases (OL-s, OL-o). In the self conditions (IL, AO-s, OL-s), the agent chooses an arm and observes its own reward. In the observation conditions, the agent receives partial social information only: AO-o reveals the demonstrator’s action, whereas OL-o reveals both the demonstrator’s action and reward. This design allows us to compare how action-only versus action–outcome social information affects subsequent self-choice on the same stimulus pair.

B. GAINS Architecture

GAINS, or **G**ated **A**rbitration of **I**ndividual and **S**ocial **L**earNing **S**ignals (Fig. 2) is a meta-learning architecture that learns to adaptively arbitrate between self-experienced and social evidence based on the task context. At each trial, the left and right bandit images are encoded into feature vectors by a shared visual encoder:

$$f_{t,i} = f_{\theta}(x_{t,i}), \quad i \in \{L, R\}. \quad (1)$$

The model maintains a causal episodic memory of past within-session events. For each completed trial $j < t$, let $a_j \in \{L, R\}$ denote the chosen arm, $u_j \in \{0, 1\}$ the actor identity (self = 0, demonstrator = 1), and $r_j \in \{0, 1, 2\}$ the reward code, where $r_j = 2$ represents the no_feedback state.

The feature representation of the chosen bandit is

$$\tilde{f}_j = \mathbb{I}[a_j = L] f_{j,L} + \mathbb{I}[a_j = R] f_{j,R}. \quad (2)$$

Each event is stored as a key–value pair. The key is obtained by a learned projection of the chosen-bandit feature,

$$K_j = q_{\phi}(\tilde{f}_j), \quad (3)$$

and the value is formed by summing learned embeddings of the action, actor, and reward:

$$V_j = e_{\phi_{\text{act}}}^{\text{act}}(a_j) + e_{\phi_{\text{actor}}}^{\text{actor}}(u_j) + e_{\phi_{\text{rwd}}}^{\text{rwd}}(r_j). \quad (4)$$

The keys and values are used to generate an episodic memory given by

$$K_{<t} = [K_j : j < t], \quad V_{<t} = [V_j : j < t]. \quad (5)$$

C. Condition-Specific Context Retrieval

To retrieve the relevant memories from the episodic memory in Equation 5, we define query for option $i \in \{L, R\}$ as

$$q_{t,i} = q_{\phi}(f_{t,i}), \quad (6)$$

where $q_{\phi}(\cdot)$ from Equation 3 is reused, thereby enforcing a shared representation for retrieval. For a past event $j < t$, let

$$\tau_j \in \{\text{IL}, \text{AO-s}, \text{AO-o}, \text{OL-s}, \text{OL-o}\}$$

denote its condition label. GAINS extracts condition-specific memory events related to each of the conditions separately in τ_j given by

$$K_{<t}^{(m)} = [K_j : j < t, \tau_j = m], \quad (7)$$

$$V_{<t}^{(m)} = [V_j : j < t, \tau_j = m]. \quad (8)$$

In Fig. 2, mask_{IL} , $\text{mask}_{\text{AO-s}}$, $\text{mask}_{\text{AO-o}}$, $\text{mask}_{\text{OL-s}}$, and $\text{mask}_{\text{OL-o}}$ denote fixed condition filters over episodic memory.

GAINS then forms a condition-specific context vector by attending only to memory tokens from condition m :

$$c_{t,i}^{(m)} = \text{LN}\left(q_{t,i} + \text{MHA}\left(q_{t,i}, K_{<t}^{(m)}, V_{<t}^{(m)}\right)\right), \quad (9)$$

where MHA is multi-head attention over masked memory and LN is layer normalization. Thus, $c_{t,i}^{(m)}$ captures the past memory entries from condition m most relevant to $q_{t,i}$.

For each attention head h , attention is computed only over past keys with $\tau_j = m$:

$$\alpha_{t,i,j}^{(h,m)} = \frac{\exp\left(\frac{(W_Q^{(h)} q_{t,i})^\top (W_K^{(h)} K_j)}{\sqrt{d_h}}\right)}{\sum_{j' < t, \tau_{j'}=m} \exp\left(\frac{(W_Q^{(h)} q_{t,i})^\top (W_K^{(h)} K_{j'})}{\sqrt{d_h}}\right)}, \quad j < t, \quad (10)$$

where $W_Q^{(h)}$, $W_K^{(h)}$, and $W_V^{(h)}$ are the learned query, key, and value projection matrices for head h , and d_h is the dimensionality of that head. The head output is the corresponding weighted sum of projected values:

$$o_{t,i}^{(h,m)} = \sum_{j < t, \tau_j=m} \alpha_{t,i,j}^{(h,m)} W_V^{(h)} V_j. \quad (11)$$

Thus, each head aggregates the most relevant past value vectors from the same condition.

We further manipulated memory accessibility through a retention parameter $\lambda_{\text{mem}} \in (0, 1]$, which exponentially discounts older memory tokens during retrieval. For a query at trial t , each past event $j < t$ is assigned age

$$a_{t,j} = t - j - 1, \quad (12)$$

so that the most recent past event has age 0. Equation 10 is then updated to

$$\alpha_{t,i,j}^{(h,m)} = \frac{\lambda_{\text{mem}}^{a_{t,j}} \exp\left(\frac{(W_Q^{(h)} q_{t,i})^\top (W_K^{(h)} K_j)}{\sqrt{d_h}}\right)}{\sum_{j' < t, \tau_{j'}=m} \lambda_{\text{mem}}^{a_{t,j'}} \exp\left(\frac{(W_Q^{(h)} q_{t,i})^\top (W_K^{(h)} K_{j'})}{\sqrt{d_h}}\right)}. \quad (13)$$

so retrieval remains condition-specific while favoring more recent memory items.

D. Self and Social Value Heads

To disentangle value estimates based on self-experience from those based on social evidence, we use three separate prediction heads. All self-choice trials where current trial condition $m_t \in \{\text{IL}, \text{AO-s}, \text{OL-s}\}$, share a common head $h_{\phi_{\text{self}}}^{\text{self}}$, whereas the two observation conditions use distinct social heads, $h_{\phi_{\text{AO}}}^{\text{AO}}$ and $h_{\phi_{\text{OL}}}^{\text{OL}}$.

For self-choice trials, we first define a condition-dependent self context by selecting the appropriate memory summary:

$$c_{t,i}^{(\text{self})} = c_{t,i}^{(\text{IL})} \mathbf{1}[m_t = \text{IL}] + c_{t,i}^{(\text{AO-s})} \mathbf{1}[m_t = \text{AO-s}] + c_{t,i}^{(\text{OL-s})} \mathbf{1}[m_t = \text{OL-s}]. \quad (14)$$

The corresponding reward logits are then

$$\ell_{t,i}^{\text{self}} = h_{\phi_{\text{self}}}^{\text{self}}([f_{t,i}; c_{t,i}^{(\text{self})}]), \quad (15)$$

For self-choice trials where $m_t \in \{\text{AO-s}, \text{OL-s}\}$, the corresponding social reward logits are computed from the matched observation streams:

$$\ell_{t,i}^{\text{AO}} = h_{\phi_{\text{AO}}}^{\text{AO}}([f_{t,i}; c_{t,i}^{(\text{AO-o})}]) \quad (16)$$

$$\ell_{t,i}^{\text{OL}} = h_{\phi_{\text{OL}}}^{\text{OL}}([f_{t,i}; c_{t,i}^{(\text{OL-o})}]). \quad (17)$$

Here, $h_{\phi_{\text{self}}}^{\text{self}}$, $h_{\phi_{\text{AO}}}^{\text{AO}}$, and $h_{\phi_{\text{OL}}}^{\text{OL}}$ denote learned prediction heads for self-experience, action-only observation, and action-outcome observation, respectively, with parameters ϕ_{self} , ϕ_{AO} , and ϕ_{OL} . The resulting arm-wise reward probabilities are obtained by applying a sigmoid to each logit:

$$p_{t,i}^{\text{self}} = \sigma(\ell_{t,i}^{\text{self}}), \quad p_{t,i}^{\text{AO}} = \sigma(\ell_{t,i}^{\text{AO}}), \quad p_{t,i}^{\text{OL}} = \sigma(\ell_{t,i}^{\text{OL}}). \quad (18)$$

Here, $p_{t,i}^{\text{self}}$ denotes the model's belief about choice $i \in \{L, R\}$ based on self-choice trials at trial t ; $p_{t,i}^{\text{AO}}$ denotes the action-based social preference for choice i inferred from previous AO-o trials; and $p_{t,i}^{\text{OL}}$ denotes the social belief about bandit i inferred from prior OL-o trials.

E. Social Arbitration

GAINS uses separate gating functions to control the influence of action-based and outcome-based social information:

$$g_{\text{AO},t} = \sigma\left(h_{\phi_{g,\text{AO}}}^{g,\text{AO}}([c_{t,L}^{\text{AO-s}}, c_{t,R}^{\text{AO-s}}, c_{t,L}^{\text{AO-o}}, c_{t,R}^{\text{AO-o}}])\right), \quad (19)$$

$$g_{\text{OL},t} = \sigma\left(h_{\phi_{g,\text{OL}}}^{g,\text{OL}}([c_{t,L}^{\text{OL-s}}, c_{t,R}^{\text{OL-s}}, c_{t,L}^{\text{OL-o}}, c_{t,R}^{\text{OL-o}}])\right). \quad (20)$$

where $h_{\phi_{g,\text{AO}}}^{g,\text{AO}}$ and $h_{\phi_{g,\text{OL}}}^{g,\text{OL}}$ are learned gating networks parameterized by $\phi_{g,\text{AO}}$ and $\phi_{g,\text{OL}}$.

The scalars $g_{\text{AO},t}$ and $g_{\text{OL},t}$ determine how strongly action-based and outcome-based social information, respectively, influence the final armwise belief. The final belief is computed as

$$p_{t,i}^{\text{belief}} = \begin{cases} p_{t,i}^{\text{self}}, & \text{if } m_t = \text{IL}, \\ (1 - g_{\text{AO},t}) p_{t,i}^{\text{self}} + g_{\text{AO},t} p_{t,i}^{\text{AO}}, & \text{if } m_t = \text{AO-s}, \\ (1 - g_{\text{OL},t}) p_{t,i}^{\text{self}} + g_{\text{OL},t} p_{t,i}^{\text{OL}}, & \text{if } m_t = \text{OL-s}. \end{cases} \quad (21)$$

F. Policy and Value Prediction

The policy and value predictions are generated from a control feature summary

$$z_t = [f_{t,L}, f_{t,R}, p_{t,L}^{\text{belief}}, p_{t,R}^{\text{belief}}]. \quad (22)$$

The final policy and value outputs are

$$\pi_t = h_{\phi_{\pi}}(z_t), \quad V_t = h_{\phi_V}(z_t). \quad (23)$$

where $h_{\phi_{\pi}}$ and h_{ϕ_V} are learned value and policy heads parameterized by ϕ_{π} and ϕ_V respectively.

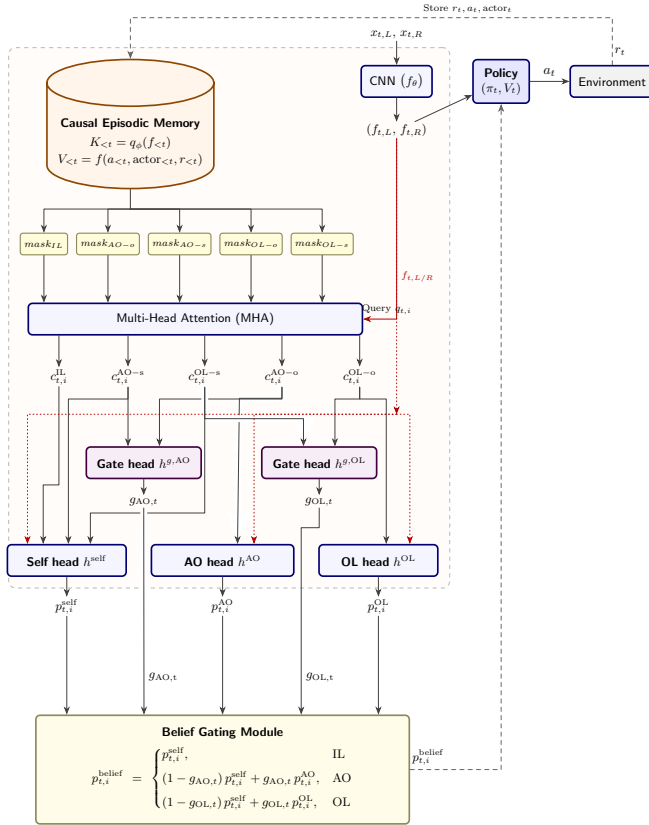


Fig. 2. GAINS: A shared visual encoder and causal episodic memory support three learning streams: Individual Learning (IL), Action-Only (AO), and Observational Learning (OL). IL relies on self-experience only, AO combines self-experience with observed demonstrator actions, and OL combines self-experience with observed demonstrator actions and outcomes. Separate gates g_{AO} and g_{OL} control how strongly each social stream influences choice.

G. Training Objective

GAINS is optimized using a hybrid objective that combines PPO-based reinforcement learning on self-choice trials with supervised belief learning. The PPO objective trains the agent to meta-learn an internal reinforcement learning procedure that maximizes episodic return in each session with novel stimulus pairs and unknown demonstrator and task reward probabilities. In parallel, a supervised self-belief loss trains the belief module, including the arbitration gates g_{AO} and g_{OL} , which dynamically balance self-experienced and social evidence based on the inferred context. The belief and policy modules are optimized separately, such that gradients from the policy objective are not propagated into the belief module.

Let t_{self} denote the index of a self-choice time step within a session. At each such time step, the agent observes the feature summary $z_{t_{\text{self}}}$, samples an action $a_{t_{\text{self}}}$, and receives reward $r_{t_{\text{self}}}$. Using generalized advantage estimation (GAE),

the policy is optimized with

$$\mathcal{L}_{\text{PPO}} = -\mathbb{E}_{t_{\text{self}}} \left[\min \left(\omega_{t_{\text{self}}}(\theta) \hat{A}_{t_{\text{self}}}, \text{clip}(\omega_{t_{\text{self}}}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_{t_{\text{self}}} \right) \right], \quad (24)$$

where

$$\omega_{t_{\text{self}}}(\theta) = \frac{\pi_{\theta}(a_{t_{\text{self}}} | z_{t_{\text{self}}})}{\pi_{\theta_{\text{old}}}(a_{t_{\text{self}}} | z_{t_{\text{self}}})}, \quad (25)$$

and the advantage estimate $\hat{A}_{t_{\text{self}}}$ is computed using GAE:

$$\hat{A}_{t_{\text{self}}} = \sum_{l=0}^{T_{\text{self}} - t_{\text{self}} - 1} (\gamma \lambda_{\text{GAE}})^l \delta_{t_{\text{self}} + l}, \quad (26)$$

with temporal-difference residual

$$\delta_{t_{\text{self}}} = r_{t_{\text{self}}} + \gamma V(z_{t_{\text{self}}+1}) - V(z_{t_{\text{self}}}). \quad (27)$$

Here, T_{self} denotes the number of self-choice time steps in the session. The value head is trained with a squared-error loss

$$\mathcal{L}_{\text{value}} = \mathbb{E}_{t_{\text{self}}} \left[(V(z_{t_{\text{self}}}) - \hat{R}_{t_{\text{self}}})^2 \right], \quad (28)$$

where $\hat{R}_{t_{\text{self}}}$ denotes the bootstrapped return.

In addition, GAINS is trained with a supervised self-belief loss on self-choice time steps with observed feedback. Let $\mathcal{S}$ denote the set of such time steps, and let $\ell_{t_{\text{self}}, a_{t_{\text{self}}}}^{\text{belief}}$ denote the fused belief logit assigned to the action actually selected by the agent at self-choice time step t_{self} , obtained by applying the logit function to the fused belief in Equation 21. This logit is trained to predict the realized reward $r_{t_{\text{self}}} \in \{0, 1\}$ using binary cross-entropy:

$$\mathcal{L}_{\text{self}} = -\frac{1}{|\mathcal{S}|} \sum_{t_{\text{self}} \in \mathcal{S}} \left[r_{t_{\text{self}}} \log \sigma(\ell_{t_{\text{self}}, a_{t_{\text{self}}}}^{\text{belief}}) + (1 - r_{t_{\text{self}}}) \log \left(1 - \sigma(\ell_{t_{\text{self}}, a_{t_{\text{self}}}}^{\text{belief}}) \right) \right], \quad (29)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ denotes the sigmoid function. Intuitively, $\mathcal{L}_{\text{self}}$ encourages the agent's internal belief about its chosen action to align with the realized reward outcome from its own experience.

The overall bandit objective is then

$$\mathcal{L}_{\text{bandit}} = \mathcal{L}_{\text{PPO}} + c_v \mathcal{L}_{\text{value}} - c_e \mathcal{H} + c_{\text{self}} \mathcal{L}_{\text{self}}, \quad (30)$$

where $\mathcal{H}$ is the policy entropy term, and c_v , c_e , and c_{self} are weighting coefficients controlling the contributions of value regression, entropy regularization, and supervised self-belief learning, respectively.

III. RESULTS

A. Evaluation Setup and Metrics

We trained and evaluated GAINS across 15 social learning scenarios, defined by the combination of three demonstrator regimes and five bandit difficulty settings. The demonstrator regimes were *expert* (choosing the higher-reward option with probability 0.9), *learning* (Q-learning; $\alpha = 0.28$, $\beta = 0.3$), and *unreliable* (random choice). Bandit difficulty

was manipulated by fixing $p_{hi} = 0.8$ and varying $p_{lo} \in \{0.2, 0.3, 0.4, 0.5, 0.6\}$. Each training update used 16 sessions per condition, and training ran for up to 1500 updates with Adam at a learning rate of 10^{-3} . To investigate how memory accessibility shapes social learning, we repeated training and evaluation at three levels of memory decay, $\lambda_{mem} \in \{1.0, 0.6, 0.05\}$.

Each session contained three independent stimulus pairs corresponding to IL, AO, and OL. Each pair was presented 30 times, yielding 90 self-choice trials per session. In addition, 30 observation trials were included for the AO and OL conditions (AO-o and OL-o), using the same stimuli as their corresponding self-choice trials (AO-s and OL-s). Following [24], each observation trial preceded its corresponding self-experience trial, though not necessarily on consecutive trials. Observations consisted of RGB frames at 384×400 pixels, with the two stimuli randomly assigned to the left and right sides across trials.

We used this setup to address two main questions: first, how demonstrator regime and task difficulty modulate the benefit of social learning; and second, how reduced memory accessibility affects both absolute performance and the relative value of social information.

B. Social Benefit Depends on Demonstrator Mode and Task Difficulty

The value of social information under full memory depended on demonstrator mode and task difficulty (Table I). With expert demonstrators, AO and OL both outperformed IL. With learning demonstrators, OL remained superior while AO offered little gain over IL. With unreliable demonstrators, AO fell below IL, but OL still exceeded it. Thus, outcome-based social information can be useful even with suboptimal demonstrators, whereas action-only information is more fragile.

TABLE I

PERFORMANCES AT $\lambda_{mem} = 1.0$ BY DEMONSTRATOR MODE AND TASK DIFFICULTY. VALUES ARE MEAN $\pm$ SD OF HIGH-ARM PICK RATE (%). EACH OF THE 15 MODE $\times$ DIFFICULTY CONDITIONS WAS ESTIMATED FROM 50 EVALUATION SESSIONS. DEMONSTRATOR-MODE ROWS AVERAGE ACROSS FIVE REWARD SETTINGS; TASK-DIFFICULTY ROWS AVERAGE ACROSS THREE DEMONSTRATOR REGIMES.

Mode	IL	AO	OL
Expert	82.25 $\pm$ 7.18	90.27 $\pm$ 2.29	89.59 $\pm$ 2.44
Learning	80.59 $\pm$ 8.73	81.00 $\pm$ 8.14	84.03 $\pm$ 8.18
Unreliable	81.33 $\pm$ 6.61	79.84 $\pm$ 7.86	86.87 $\pm$ 7.97
<i>Task difficulty averages</i>			
p_{lo}	IL	AO	OL
0.2	88.62 $\pm$ 1.22	90.53 $\pm$ 2.31	91.71 $\pm$ 1.71
0.3	85.47 $\pm$ 0.53	88.02 $\pm$ 3.48	91.09 $\pm$ 1.80
0.4	84.04 $\pm$ 1.26	84.31 $\pm$ 5.88	89.16 $\pm$ 1.39
0.5	79.24 $\pm$ 1.42	79.60 $\pm$ 8.15	85.16 $\pm$ 2.30
0.6	69.58 $\pm$ 3.28	76.04 $\pm$ 9.98	77.02 $\pm$ 8.84

Task difficulty showed a similarly selective effect (Table I). OL advantage over IL increased from 3.09 points at

$p_{lo} = 0.2$ to 7.44 points at $p_{lo} = 0.6$, indicating that outcome-based social information became relatively more helpful as the task grew harder. By contrast, AO gains over IL were smaller and less regular across difficulty levels. Overall the clearest social-learning benefit arose when the agent observed both demonstrator actions and outcomes (OL-o), whereas action-only social evidence (AO-o) produced weaker and less consistent gains across different task settings.

C. Memory Decay Lowers Absolute Performance but Increases the Relative Value of Social Information

We next examined how reducing memory accessibility altered performance (Fig. 3, Table II). As λ_{mem} decreased from 1.0 to 0.6 and 0.05, absolute performance declined in all three conditions. Mean high-arm pick rate changed from 81.39, 83.70, and 86.83 at $\lambda_{mem} = 1.0$ to 71.50, 77.35, and 79.87 at $\lambda_{mem} = 0.6$, and to 65.54, 73.08, and 74.92 at $\lambda_{mem} = 0.05$, for IL, AO, and OL respectively. However, OL still outperformed both AO and IL across all three λ_{mem} settings.

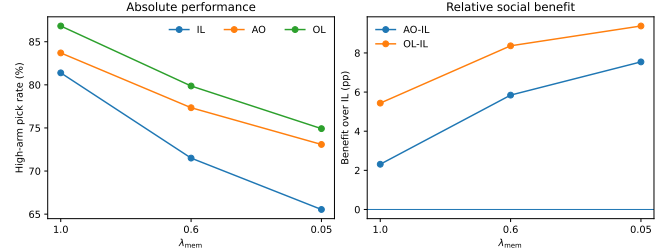


Fig. 3. Memory decay reduces absolute performance but increases the relative value of social information. Left: mean high-arm pick rate for IL, AO, and OL across $\lambda_{mem} \in \{1.0, 0.6, 0.05\}$. Right: relative social benefit, measured as AO-IL and OL-IL, across the same settings.

However, this decline was largest for IL. As a result, the relative benefit of social information increased as memory weakened. The AO advantage over IL grew from 2.31 points at $\lambda_{mem} = 1.0$ to 5.84 at $\lambda_{mem} = 0.6$ and 7.54 at $\lambda_{mem} = 0.05$. The OL advantage over IL increased from 5.44 to 8.36 and 9.38, respectively. Thus, reduced access to past self-experience lowered overall performance, but also made social information comparatively more beneficial in this task.

D. Gate Dynamics Reveal Rapid Suppression When Action-Only Social Evidence Is Noisy

To test gate dynamics, we set $\lambda_{mem} = 1.0$. The AO gate $g_{AO,t}$ decreased fastest for unreliable demonstrators, more gradually for learning demonstrators, and slowest for experts (Fig. 4).

The OL gate $g_{OL,t}$ followed a different pattern: it started high and stayed high for expert demonstrators, indicating that outcome-based social evidence was useful from the outset. For learning and unreliable demonstrators, $g_{OL,t}$ rose more gradually and stabilized later than $g_{AO,t}$. Together, these dynamics suggest that GAINS quickly down-weights noisy AO evidence while continuing to integrate richer OL evidence even when social evidence is noisy.

TABLE II

PAIRED CONTRASTS IN HIGH-ARM PICK RATE ACROSS MEMORY SETTINGS. RAW p -VALUES ARE FROM TWO-SIDED ONE-SAMPLE t -TESTS ON PAIRED DIFFERENCES ACROSS CONDITION $\times$ DIFFICULTY COMBINATIONS. BENJAMINI-HOCHBERG CORRECTED p -VALUES WERE COMPUTED *within each memory setting* ACROSS THE THREE PLANNED CONTRASTS REPORTED FOR THAT SETTING.

λ_{mem}	Contrast	Mean diff.	95% CI	p	p_{BH}	Sig. after BH
1.00	AO-IL	2.31	[-0.97, 5.59]	0.153	0.153	No
1.00	OL-IL	5.44	[3.49, 7.38]	< 0.001	< 0.001	Yes
1.00	OL-AO	3.12	[0.73, 5.52]	0.014	0.022	Yes
0.60	AO-IL	5.84	[-0.69, 12.38]	0.076	0.076	No
0.60	OL-IL	8.36	[3.49, 13.24]	0.002	0.007	Yes
0.60	OL-AO	2.52	[-0.02, 5.06]	0.052	0.076	No
0.05	AO-IL	7.54	[1.10, 13.99]	0.025	0.037	Yes
0.05	OL-IL	9.38	[4.96, 13.79]	< 0.001	0.001	Yes
0.05	OL-AO	1.84	[-0.89, 4.56]	0.171	0.171	No

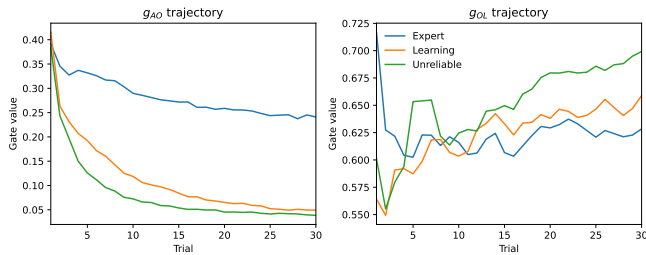


Fig. 4. Average within-session gate trajectories under full memory access ($\lambda_{\text{mem}} = 1.0$), averaged across the five reward settings within each demonstrator mode. The AO gate is rapidly suppressed for unreliable and learning demonstrators, whereas the OL gate remains high for experts and rises more gradually for learning and unreliable demonstrators.

E. Effect of fixed-gate ablations on observational performance

We compared the learned-gate agent ($\lambda_{\text{mem}} = 1.0$) with fixed-gate ablations ($g = 0.5$, $g = 0.7$; Table III) using linear models of high-arm pick rate: ($\text{pickuprate} \sim \text{gate} + \text{difficulty}$), where *gate* denotes the agent type (learned-g, $g = 0.5$, $g = 0.7$) and *difficulty* denotes task difficulty (p_{lo}). Each model was fit separately for each observation condition (AO, OL) and demonstrator mode, using 3000 session-level observations (3 gate settings $\times$ 5 difficulty levels $\times$ 200 sessions).

Fixed gates most strongly hurt performance under noisy demonstrators: both ablations significantly reduced pick rate in AO, while only ($g = 0.5$) did so in OL. Under learning demonstrators, ($g = 0.5$) significantly reduced performance in both conditions; ($g = 0.7$) did not. Under expert demonstrators the effects were mixed, with ($g = 0.5$) improving AO and ($g = 0.7$) reducing OL. These results suggest adaptive gating is most beneficial when demonstrator information is noisy.

IV. DISCUSSION

We introduced GAINS, a social learning model that arbitrates between self-experience and socially observed evidence. Overall, social information improved learning, but its

TABLE III

MEAN CHANGE IN HIGH-ARM PICK RATE (Δ , PERCENTAGE POINTS) FOR FIXED-GATE ABLATIONS ($g = 0.5$, $g = 0.7$) RELATIVE TO THE LEARNED-GATE AGENT ($\lambda_{\text{mem}} = 1.0$), BY DEMONSTRATOR TYPE AND OBSERVATION CONDITION (AO, OL). p_{BH} : BENJAMINI-HOCHBERG CORRECTED p -VALUES FROM LINEAR MODELS ($\text{pickuprate} \sim \text{gate} + \text{difficulty}$).

Demonstrator	Cond.	$g = 0.5$		$g = 0.7$	
		Δ (%)	p_{BH}	Δ (%)	p_{BH}
Expert	AO	3.88	< 10^{-4}	0.95	0.1713
Expert	OL	-1.12	0.0672	-3.36	< 10^{-4}
Unreliable	AO	-8.04	< 10^{-4}	-4.39	< 10^{-4}
Unreliable	OL	-1.98	0.0022	-1.14	0.1289
Learning	AO	-2.12	0.0206	-1.42	0.1318
Learning	OL	-2.36	0.0020	-1.39	0.0792

value depended on both informational content and demonstrator reliability. Observational Learning (OL), providing both actions and outcomes, yielded the strongest and most robust gains over Individual Learning across all conditions. For expert demonstrators, Action-Only (AO) observation performed comparably to OL, since reliable actions alone carry a strong learning signal. For unreliable demonstrators, however, OL outperformed AO and IL, because the observed outcomes helped assess whether actions were informative. In the two-choice bandit setting, OL-o observations provide an informational advantage comparable to direct experience. Rather than being explicitly programmed to exploit this structure, GAINS learns to use this advantage across conditions. This contrasts with human social learning, where learners often selectively discount information from unreliable sources [26], [19].

Reducing memory access impaired performance across the board, but also increased the relative benefit of social information, particularly under OL. This suggests that socially observed evidence can partially compensate for limited self-experience, and we propose memory accessibility as a candidate mechanism underlying developmental differences

in social-versus-individual learning arbitration [25].

Gate dynamics support interpreting GAINS as adaptive arbitration rather than imitation. AO evidence was rapidly down-weighted for noisy demonstrators, while outcome-based evidence remained engaged, indicating that GAINS learns to weight social information by both source reliability and informational richness, not by a simple demonstrator-reliability filter. Fixed-gate ablations confirmed this: replacing learned gates with fixed values hurt performance most under unreliable demonstrators and in the AO condition, showing that GAINS's benefit comes from adaptive modulation, not mere access to social information.

In short, the usefulness of social learning depends on what is observed, who is observed, and how much the learner can draw on its own past experience. Future work should extend these findings to richer environments, multiple demonstrators, and direct comparisons with human learners under developmental or memory constraints.

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